The Period-4 Morton–Vivaldi Irreducibility Conjecture and Abelian Stability

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Declaration on AI assistance. The core irreducibility proof in this paper was obtained using OpenAI Sol Pro.

Abstract

Let Φ_k denote the k-th cyclotomic polynomial, let φ denote Euler's totient function, and put

$$g(T) = -T^4 - 2T^3 - 4T^2 - 6T + 5 + \frac{8}{T} + \frac{16}{T^2}.$$

We prove, for every $k \geq 2$, that

$$\Gamma_k(T) = (-1)^{\varphi(k)} T^{2\varphi(k)} \Phi_k(g(T))$$

is irreducible over $\mathbf{Q}$. This settles the complete period-4 case of the Morton–Vivaldi irreducibility conjecture for the quadratic family. We then prove the stronger fiberwise statement that, for every root of unity $\zeta \neq 1$, the sextic F_ζ remains irreducible over the maximal abelian extension of $\mathbf{Q}(\zeta)$. The resulting theorem gives the complete factorization of the pullbacks and of the period-4 delta factors over every finite abelian extension of $\mathbf{Q}$, as well as the maximal abelian subfield of each parameter field. A central ingredient is an abstract criterion: an irreducible sextic admitting a local irreducible quartic factor and whose $3+3$ resolvent has no ground-field root is disjoint from the maximal abelian extension of its ground field.

1 Introduction

For the quadratic family $f_c(z) = z^2 + c$, Morton and Vivaldi introduced delta factors whose roots are the parabolic parameters of prescribed formal and exact periods, and conjectured their irreducibility over $\mathbf{Q}$ [MV95]. Morton subsequently restated the irreducibility problem [Mor96, Introduction]; it also appears as an open problem in [Sil07, Exercise 4.12(e)**]. The cases of periods 1 and 2 reduce to cyclotomic irreducibility [MV95, Corollary 3.7], and the complete period-3 case was proved by Koizumi, Murakami, Sano, and Takehira [KMST25, Theorem 1.2]. The first remaining complete period is therefore period 4.

Put $C = 4c$. The period-4 multiplier curve has normalization

$$T \longmapsto \big(g(T), h(T)\big), \qquad h(T) = -T^2 - 3 - \frac{4}{T}, \tag{1}$$

on $\mathbf{G}_m$. This parametrization is recorded in [KMST25, Remark 2.10] and attributed there to Giarrusso and Fisher [GF95]. In Section 8 we give a direct algebraic proof that the displayed map is the normalization. Pulling cyclotomic polynomials back along the degree-six function g leads, for $k \geq 2$, to

$$\Gamma_k(T) = (-1)^{\varphi(k)} T^{2\varphi(k)} \Phi_k(g(T)).$$

This Laurent expression is in fact a monic polynomial in $\mathbf{Z}[T]$ of degree $6\varphi(k)$.

Our main result is the following.

Theorem 1.1. *For every integer $k \geq 2$, the polynomial $\Gamma_k(T)$ is irreducible over $\mathbf{Q}$.*

The second main theorem strengthens the corresponding fixed-degree statement. For a number field K, write K^{ab} for its maximal abelian extension in a fixed algebraic closure. For a root of unity ζ, set

$$F_\zeta(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (\zeta - 5)T^2 - 8T - 16.$$

Theorem 1.2 (Abelian-stable irreducibility)**.** *Let $\zeta \neq 1$ be any root of unity and put $K = \mathbf{Q}(\zeta)$. Then*

$$F_\zeta(T) \text{ is irreducible over } K^{\mathrm{ab}}.$$

Equivalently, if α is a root of F_ζ, then

$$K(\alpha) \cap K^{\mathrm{ab}} = K.$$

The normalization of the multiplier curve transfers the first theorem to the period-4 delta factors.

Corollary 1.3. *For every integer $k \geq 1$, the period-4 delta factor $\Delta_{4k,4}$ is irreducible over $\mathbf{Q}$.*

The proof of Theorem 1.1 combines reduction at primes above 2 with a uniform archimedean root bound. For Theorem 1.2, we prove an abstract abelian-disjointness criterion for irreducible sextics. Its two inputs are an irreducible quartic factor over one completion and the absence of a ground-field root of a $3+3$ resolvent. Newton polygons supply the local quartic factors for both the sextic and its explicit resolvent. The resolvent identity and the three polynomial congruences used in its local analysis are accompanied by a deterministic exact-integer certificate in Appendix A.

2 The fixed-degree reformulation

Let ζ be a root of unity different from 1, and recall

$$F_\zeta(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (\zeta - 5)T^2 - 8T - 16. \tag{2}$$

A direct calculation gives

$$g(T) - \zeta = -\frac{F_\zeta(T)}{T^2}. \tag{3}$$

Proposition 2.1. *Let $k \geq 2$, fix a primitive k-th root ζ_0, and put $K = \mathbf{Q}(\zeta_0)$. The following conditions are equivalent:*

(A) *Γ_k is irreducible over $\mathbf{Q}$;*

(B) *F_{ζ_0} is irreducible over K;*

(C) *$F_{\zeta'}$ is irreducible over K for every primitive k-th root ζ'.*

Proof. The cyclotomic factorization and (3) give

$$\begin{aligned}\Gamma_k(T) &= (-1)^{\varphi(k)}T^{2\varphi(k)} \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} (g(T)-\zeta) \\ &= \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} F_\zeta(T).\end{aligned}$$

Thus Γ_k is monic of degree $6\varphi(k)$. Its coefficients are algebraic integers fixed by the Galois group of $K/\mathbf{Q}$, so they belong to $\mathbf{Z}$.

Let α be a root of F_{ζ_0}. Since $F_{\zeta_0}(0) = -16$, we have $\alpha \neq 0$, and (3) yields

$$\zeta_0 = g(\alpha) \in \mathbf{Q}(\alpha).$$

Consequently $K \subseteq \mathbf{Q}(\alpha)$. If F_{ζ_0} is irreducible over K, then

$$[\mathbf{Q}(\alpha) : \mathbf{Q}] = [K(\alpha) : K][K : \mathbf{Q}] = 6\varphi(k).$$

The minimal polynomial of α over $\mathbf{Q}$ therefore has the same degree as Γ_k and divides it, so it equals Γ_k.

Conversely, suppose that Γ_k is irreducible. Then every root α of F_{ζ_0} has degree $6\varphi(k)$ over $\mathbf{Q}$. Since $K \subseteq \mathbf{Q}(\alpha)$, it follows that $[K(\alpha) : K] = 6$. The minimal polynomial of α over K has degree 6 and divides F_{ζ_0}, so F_{ζ_0} is irreducible. Finally, the automorphisms of $K/\mathbf{Q}$ act transitively on the primitive k-th roots and send F_{ζ_0} to the corresponding polynomials $F_{\zeta'}$. This proves all three equivalences. □

At $\zeta = 1$ one has the exceptional factorization

$$F_1(T) = (T+2)(T^2+4)(T^3-2). \tag{4}$$

Set

$$P(T) = F_1(T).$$

Then

$$F_\zeta(T) = P(T) + (\zeta - 1)T^2. \tag{5}$$

3 A uniform archimedean bound

Lemma 3.1. *Let $|\zeta| = 1$. Every complex root α of F_ζ satisfies*

$$1 < |\alpha| < \frac{5}{2}.$$

Proof. Put $r = |\alpha|$. By (5),

$$|P(\alpha)| = |\zeta - 1|r^2 \leq 2r^2. \tag{6}$$

If $r \leq 1$, then the reverse triangle inequality and (4) give

$$|P(\alpha)| \geq (2-r)(4-r^2)(2-r^3) \geq 3,$$

whereas $2r^2 \leq 2$, contradicting (6).

Now suppose that $r \geq 5/2$. Again by the reverse triangle inequality,

$$\frac{|P(\alpha)|}{r^2} \geq (r-2)(r^2-4)\left(r - \frac{2}{r^2}\right).$$

Each of the three factors on the right is positive and increasing for $r \geq 5/2$. Hence

$$\frac{|P(\alpha)|}{r^2} \geq \frac{1}{2} \cdot \frac{9}{4} \cdot \frac{109}{50} = \frac{981}{400} > 2,$$

which also contradicts (6). The claimed strict bounds follow. $\square$

We shall use both sides of the annular bound. If $K = \mathbf{Q}(\zeta)$, then the constant term of any positive-degree monic factor of F_ζ in $K[T]$ has absolute value greater than 1 under every embedding of K into $\mathbf{C}$, while each individual root has absolute value less than $5/2$.

4 Two factorization lemmas

Let K be a number field, let $\mathcal{O}_K$ be its ring of integers, and let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$. Write $v_\mathfrak{p}$ for the additive valuation normalized by $v_\mathfrak{p}(K^\times) = \mathbf{Z}$. A bar denotes reduction modulo $\mathfrak{p}$.

Lemma 4.1. *If a monic polynomial $f \in \mathcal{O}_K[T]$ factors as $f = HJ$ with H, J monic in $K[T]$, then $H, J \in \mathcal{O}_K[T]$.*

Proof. Every root of f is integral over $\mathcal{O}_K$. The coefficients of H and J are elementary symmetric functions in subsets of these roots, hence are integral over $\mathcal{O}_K$. They also lie in K. Since $\mathcal{O}_K$ is integrally closed in K, they belong to $\mathcal{O}_K$. $\square$

Lemma 4.2 (the unsplit-block lemma)**.** *Let $f \in \mathcal{O}_K[T]$ be monic and suppose that, for some integers $a, b \geq 0$ and some $t \in \mathcal{O}_K$ with $\bar{t} \neq 0$,*

$$\bar{f}(T) = T^a (T - \bar{t})^b, \qquad f(t) \in \mathfrak{p} \setminus \mathfrak{p}^2.$$

If $f = HJ$ with $H, J \in \mathcal{O}_K[T]$ monic and $\deg H < b$*, then*

$$\bar{H}(T) = T^{\deg H} \qquad \textit{and} \qquad \deg H \leq a.$$

Proof. Unique factorization over the residue field gives integers r, s such that

$$\bar{H} = T^r (T - \bar{t})^s, \qquad \bar{J} = T^{a-r} (T - \bar{t})^{b-s}.$$

Because H and J are monic, reduction preserves their degrees. If $s > 0$, then $s \leq \deg H < b$, so both s and $b - s$ are positive. Consequently $H(t), J(t) \in \mathfrak{p}$, which would imply $f(t) = H(t)J(t) \in \mathfrak{p}^2$. This is impossible. Thus $s = 0$, so $\bar{H} = T^r$. Degree preservation gives $r = \deg H \leq a$. $\square$

We also use the standard elementary fact that if a sum in a discretely valued field is zero, then its terms cannot have a unique least valuation.

5 Cyclotomic facts at the prime 2

Let ζ have order

$$k = 2^a m, \qquad m \text{ odd},$$

and put $K = \mathbf{Q}(\zeta)$. There is a unique decomposition

$$\zeta = \xi \eta, \tag{7}$$

where ξ has order 2^a and η has order m.

We recall the local cyclotomic facts used below. If $\mathfrak{p} \mid 2$, then reduction is injective on roots of unity of odd order, while every root of unity of 2-power order reduces to 1. The odd-order cyclotomic extension is unramified at 2. If $a \geq 2$, then $\xi - 1$ is a uniformizer and

$$v_{\mathfrak{p}}(2) = 2^{a-1}. \tag{8}$$

Indeed, with $n = 2^{a-1}$, the element $\xi - 1$ is a root of

$$(1 + X)^n + 1,$$

which is Eisenstein at 2. Passing to the compositum with the unramified odd-order part preserves the uniformizer and the ramification index. When $a = 0$, the extension is unramified at 2; when $a = 1$ and m is odd, $\mathbf{Q}(\zeta_{2m}) = \mathbf{Q}(\zeta_m)$, so it is again unramified at 2. These statements are also immediate from the usual decomposition law for primes in cyclotomic fields; see, for example, [Was97, Chapter 2].

6 Orders with a nontrivial odd part

Assume throughout this section that $m > 1$. Fix a prime $\mathfrak{p} \mid 2$ and write

$$e = v_{\mathfrak{p}}(2).$$

Since 4 is invertible modulo m, there is an odd-order root of unity $\theta \in K$ such that $\theta^4 = \eta$. Put

$$t_0 = \theta - 1.$$

The reduction of θ is not 1, so t_0 is a $\mathfrak{p}$-adic unit.

Since $\bar{\xi} = 1$ and $\bar{\eta} \neq 1$, reduction of (2) in characteristic 2 gives

$$\begin{aligned} \overline{F_\zeta(T)} &= T^6 + (\bar{\eta} + 1)T^2 \\ &= T^2\big((T + 1)^4 + \bar{\eta}\big) \\ &= T^2(T - \bar{t}_0)^4. \end{aligned} \tag{9}$$

Lemma 6.1. *For every prime $\mathfrak{p} \mid 2$,*

$$v_{\mathfrak{p}}\big(F_\zeta(t_0)\big) = 1.$$

Proof. Define the Laurent polynomial

$$B(T) = T^4 + 3T^3 + 5T^2 + 5T - 2 - \frac{4}{T} - \frac{8}{T^2}.$$

The identity

$$g(T) - (T + 1)^4 = -2B(T)$$

and $(t_0 + 1)^4 = \eta$ imply

$$F_\eta(t_0) = 2t_0^2 B(t_0). \tag{10}$$

Modulo $\mathfrak{p}$,

$$\bar{B}(T) = T^4 + T^3 + T^2 + T = T(T + 1)^3.$$

This reduction is legitimate because t_0 is a $\mathfrak{p}$-adic unit. Therefore

$$\bar{B}(t_0) = \bar{t}_0\bar{\theta}^3 \neq 0,$$

and (10) gives $v_{\mathfrak{p}}(F_\eta(t_0)) = e$.

Moreover,

$$F_\zeta(T) = F_\eta(T) + \eta(\xi - 1)T^2. \tag{11}$$

If $a = 0$, then $\xi = 1$ and $e = 1$, so the assertion follows from (10). If $a \geq 2$, the two terms on the right side of (11), evaluated at t_0, have valuations $e \geq 2$ and 1, respectively, so their sum has valuation 1.

It remains to consider $a = 1$. Here $\xi = -1$ and $e = 1$, whence

$$F_\zeta(t_0) = 2t_0^2\big(B(t_0) - \eta\big).$$

After division by 2 and reduction modulo $\mathfrak{p}$,

$$\begin{aligned} \overline{\frac{F_\zeta(t_0)}{2}} &= \bar{t}_0^2\big(\bar{t}_0\bar{\theta}^3 + \bar{\theta}^4\big) \\ &= \bar{t}_0^2\bar{\theta}^3 \neq 0. \end{aligned}$$

Thus the valuation is 1 in every case. $\square$

Proposition 6.2. *If the order of ζ has a nontrivial odd part, then F_ζ is irreducible over K.*

Proof. Suppose that F_ζ is reducible over K. Choose a monic irreducible factor H of minimal degree d, and write $F_\zeta = HJ$ with J monic. Since $\deg F_\zeta = 6$, we have $d \leq 3$. By Lemma 4.1, both factors lie in $\mathcal{O}_K[T]$.

At each prime $\mathfrak{p} \mid 2$, equations (9) and Lemma 6.1 allow us to apply Lemma 4.2 with $(a, b) = (2, 4)$. Since $d < 4$, it follows that $d \leq 2$ and

$$\bar{H} = T^d. \tag{12}$$

First suppose that $d = 2$. Put

$$b = H(0), \qquad c = J(0).$$

Then $bc = -16$. At every $\mathfrak{p} \mid 2$, (12) and (9) give

$$\bar{J} = (T - \bar{t}_0)^4,$$

so c is a $\mathfrak{p}$-adic unit. At primes not above 2, the integrality of b, c and $bc = -16$ also show that c is a unit. Hence c is a unit of $\mathcal{O}_K$.

For every complex embedding $\sigma : K \hookrightarrow \mathbf{C}$, the polynomial $\sigma(J)$ is a monic factor of $F_{\sigma(\zeta)}$ of degree 4. By Lemma 3.1, each of its roots has modulus greater than 1. Thus $|\sigma(c)| > 1$ for every σ, and therefore

$$|\operatorname{N}_{K/\mathbf{Q}}(c)| = \prod_\sigma |\sigma(c)| > 1.$$

This contradicts the fact that the norm of a unit is ± 1.

Now suppose that $d = 1$, say $H(T) = T - \alpha$. From (12), $q = v_{\mathfrak{p}}(\alpha) > 0$ for every $\mathfrak{p} \mid 2$. Moreover, $\zeta - 5$ is a $\mathfrak{p}$-adic unit, because $\bar{\zeta} = \bar{\eta} \neq 1$. The valuations of the seven terms in $F_\zeta(\alpha) = 0$, ordered from the constant term to the leading term, are

$$4e, \quad 3e + q, \quad 2q, \quad e + 3q, \quad 2e + 4q, \quad e + 5q, \quad 6q.$$

If $q < 2e$, then $2q$ is the unique least value. If $q > 2e$, then $4e$ is the unique least value. Since a zero sum cannot have a unique term of least valuation, necessarily

$$v_{\mathfrak{p}}(\alpha) = 2e = v_{\mathfrak{p}}(4) \qquad (\mathfrak{p} \mid 2). \tag{13}$$

At every prime away from 2, the equality $(-\alpha)J(0) = -16$ and integrality give valuation zero for α. Thus $(\alpha) = (4)$, and $u = \alpha/4$ is a unit of $\mathcal{O}_K$.

For every complex embedding σ, Lemma 3.1 gives

$$|\sigma(u)| = \frac{|\sigma(\alpha)|}{4} < \frac{5}{8}.$$

It follows that $|\,\mathrm{N}_{K/\mathbf{Q}}(u)| < 1$, contradicting that u is a unit. Both possible degrees have been excluded, so F_ζ is irreducible. □

7 Pure powers of 2

Assume now that ζ has order 2^a with $a \geq 1$. There is a unique prime $\mathfrak{p}$ of K above 2, its residue degree is 1, and

$$e = v_{\mathfrak{p}}(2) = \varphi(2^a).$$

For $a = 1$, $K = \mathbf{Q}$ and $\zeta = -1$. For $a \geq 2$, $\zeta - 1$ is a uniformizer. In all cases,

$$v_{\mathfrak{p}}(\zeta - 5) = 1. \tag{14}$$

Indeed, for $a = 1$ one has $\zeta - 5 = -6$; for $a \geq 2$ one has $\zeta - 5 = (\zeta - 1) - 4$, whose two summands have valuations 1 and $2e > 1$.

Reduction modulo $\mathfrak{p}$ gives

$$\overline{F_\zeta(T)} = T^6. \tag{15}$$

Proposition 7.1. *If ζ has 2-power order and $\zeta \neq 1$, then F_ζ is irreducible over K.*

Proof. Suppose that F_ζ is reducible. Choose a monic irreducible factor H of minimal degree $d \leq 3$, and write $F_\zeta = HJ$ with J monic. Lemma 4.1 gives $H, J \in \mathcal{O}_K[T]$, and (15) gives

$$\bar{H} = T^d, \qquad \bar{J} = T^{6-d}. \tag{16}$$

We first exclude $d = 1$. Write $H = T - \alpha$. Then $q = v_{\mathfrak{p}}(\alpha)$ is a positive integer. The valuations of the terms of $F_\zeta(\alpha) = 0$, from constant to leading, are

$$4e, \quad 3e + q, \quad 1 + 2q, \quad e + 3q, \quad 2e + 4q, \quad e + 5q, \quad 6q.$$

If $q \leq 2e - 1$, then $1 + 2q$ is the unique least value. If $q \geq 2e$, then $4e$ is the unique least value. Both cases contradict the nonarchimedean triangle inequality. Hence $d \neq 1$.

Next suppose that $d = 3$. Write

$$H = T^3 + h_2T^2 + h_1T + h_0, \qquad J = T^3 + j_2T^2 + j_1T + j_0.$$

By (16), every displayed coefficient other than the leading coefficients belongs to $\mathfrak{p}$. The coefficient of T^2 in HJ is

$$h_2j_0 + h_1j_1 + h_0j_2 \in \mathfrak{p}^2.$$

But the coefficient of T^2 in F_ζ is $\zeta - 5$, which belongs to $\mathfrak{p} \setminus \mathfrak{p}^2$ by (14). This contradiction excludes $d = 3$.

Consequently $d = 2$. Write

$$H = T^2 + h_1 T + h_0, \qquad J = T^4 + j_3 T^3 + j_2 T^2 + j_1 T + j_0.$$

All the h_i and j_i lie in $\mathfrak{p}$. Comparing the coefficient of T^2 gives

$$\zeta - 5 = j_0 + h_1 j_1 + h_0 j_2 \equiv j_0 \pmod{\mathfrak{p}^2}. \tag{17}$$

Therefore $j_0 \in \mathfrak{p} \setminus \mathfrak{p}^2$.

Put $b = H(0)$ and $c = J(0) = j_0$. At every odd prime, the integrality of b, c and $bc = -16$ implies that c is a unit. Since $\mathfrak{p}$ is the unique prime above 2, (17) shows that

$$(c) = \mathfrak{p}.$$

The residue degree of $\mathfrak{p}$ is 1, so

$$|\,\mathrm{N}_{K/\mathbf{Q}}(c)| = \mathrm{N}(\mathfrak{p}) = 2. \tag{18}$$

For every complex embedding σ, the two roots of $\sigma(H)$ have modulus less than $5/2$ by Lemma 3.1. Since the absolute value of the product of all six roots of $F_{\sigma(\zeta)}$ is 16,

$$|\sigma(c)| = \frac{16}{|\sigma(b)|} > \frac{16}{(5/2)^2} = \frac{64}{25}.$$

Taking the product over all embeddings gives

$$|\,\mathrm{N}_{K/\mathbf{Q}}(c)| > \left(\frac{64}{25}\right)^{[K:\mathbf{Q}]} > 2,$$

contradicting (18). Thus F_ζ is irreducible. □

Theorem 7.2 (Cyclotomic-fiber irreducibility)**.** *For every root of unity $\zeta \neq 1$, the polynomial F_ζ is irreducible over $\mathbf{Q}(\zeta)$.*

Proof. Proposition 6.2 applies when the order of ζ has a nontrivial odd part, and Proposition 7.1 applies to the remaining nontrivial roots of unity. □

Theorem 1.1 now follows from Proposition 2.1.

8 The period-4 multiplier curve and the delta factors

We use the standard notation of [Mor96, MV95, KMST25]. Let $\delta_m(x, c)$ denote the multiplier polynomial for cycles of formal period m, put $C = 4c$, and set

$$\widetilde{\delta}_m(x, C) = \delta_m(x, C/4).$$

The multiplier curve X_m is the affine curve $\widetilde{\delta}_m(x, C) = 0$. A direct computation from the definition in [KMST25, Section 2] gives, in period 4,

$$\begin{aligned}\widetilde{\delta}_4(x, C) = {} & C^6 + 12C^5 + (x + 48)C^4 + (4x + 192)C^3 \\ & + (512 - 16x - x^2)C^2 + (16 - x)^3. \end{aligned} \tag{19}$$

Proposition 8.1 (Normalization of the period-4 multiplier curve)**.** *The morphism*

$$\nu : \mathbf{G}_m \longrightarrow X_4, \qquad T \longmapsto \big(g(T), h(T)\big), \qquad h(T) = -T^2 - 3 - \frac{4}{T},$$

is the normalization of X_4. In particular, it is finite, birational, and surjective.

Proof. Exact substitution in (19) gives

$$\widetilde{\delta}_4\big(g(T), h(T)\big) = 0. \tag{20}$$

This Laurent-polynomial identity is also checked coefficient by coefficient by the certificate in Appendix A.

Put

$$A = \mathbf{Q}[x, C]/\big(\widetilde{\delta}_4(x, C)\big), \qquad B = \mathbf{Q}[T, T^{-1}].$$

As a rational map $\mathbf{P}^1 \to \mathbf{P}^1$, the function

$$h(T) = -\frac{T^3 + 3T + 4}{T}$$

has degree 3. Thus

$$[\overline{\mathbf{Q}}(T) : \overline{\mathbf{Q}}(h)] = 3.$$

We claim that $g \notin \overline{\mathbf{Q}}(h)$. Otherwise $g = R(h)$ for some $R \in \overline{\mathbf{Q}}(X)$. Since g has no poles outside 0 and ∞, while every finite value of h has all its preimages in $\mathbf{G}_m$, the function R has no finite pole and is a polynomial. The pole orders at infinity force $\deg R = 2$. There the leading terms $h(T)^2 \sim T^4$ and $g(T) \sim -T^4$ force the leading coefficient of R to be -1; at zero, the leading terms $h(T)^2 \sim 16T^{-2}$ and $g(T) \sim 16T^{-2}$ force it to be 1. This contradiction proves the claim.

The degree-three extension above has no nontrivial intermediate field, so

$$\overline{\mathbf{Q}}(g, h) = \overline{\mathbf{Q}}(T).$$

Since (19) has degree three in x and vanishes at (g, h), it is the minimal polynomial of g over $\overline{\mathbf{Q}}(h)$ up to a nonzero constant. Its leading coefficient as a polynomial in x is -1, so Gauss's lemma makes it absolutely irreducible. Moreover, $g \notin \overline{\mathbf{Q}}(h)$ implies $g \notin \mathbf{Q}(h)$; since $[\mathbf{Q}(T) : \mathbf{Q}(h)] = 3$, the same prime-degree argument gives $\mathbf{Q}(g, h) = \mathbf{Q}(T)$. Consequently A is a domain, and the homomorphism $A \to B$ defined by (20) is injective and birational. In B one has

$$T^3 + (C + 3)T + 4 = 0, \qquad T^{-1} = -\frac{T^2 + C + 3}{4}.$$

Thus $B = A[T]$ is finite over A. Since B is normal, it is the integral closure of A in their common function field. Hence $\operatorname{Spec} B \to \operatorname{Spec} A$ is the normalization; being finite and dominant, it is surjective. $\square$

For $k \geq 2$, define

$$\widetilde{\Delta}_{mk,m}(C) = \Delta_{mk,m}(C/4) = \operatorname{Res}_x\big(\Phi_k(x), \widetilde{\delta}_m(x, C)\big).$$

Koizumi, Murakami, Sano, and Takehira prove that these delta factors are separable and that projection to the parameter coordinate gives the relevant Galois-equivariant root correspondence [KMST25, Proposition 2.3 and Section 2]. More precisely, if $Z(q)$ denotes the set of roots of q in $\overline{\mathbf{Q}}$, without multiplicity, then

$$\{(\lambda, C) \in X_m(\overline{\mathbf{Q}}) : \lambda \text{ is a primitive } k\text{-th root of unity}\} \xrightarrow{\sim} Z(\widetilde{\Delta}_{mk,m}). \tag{21}$$

Their degree formula, originating in [MV95, Corollary 3.3], is

$$\deg_C \widetilde{\Delta}_{mk,m} = \nu(m)\varphi(k), \qquad \nu(m) = \sum_{d|m} 2^{d-1}\mu(m/d). \tag{22}$$

Here μ is the Möbius function. In particular, $\nu(4) = 0 - 2 + 8 = 6$.

Lemma 8.2 (Root correspondence in period 4)**.** *For every $k \geq 2$, the map*

$$\beta : Z(\Gamma_k) \longrightarrow Z(\widetilde{\Delta}_{4k,4}), \qquad \beta(\alpha) = h(\alpha),$$

is a bijection equivariant for $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$.

Proof. The polynomial Γ_k is irreducible of degree $6\varphi(k)$. Its constant term is $(-16)^{\varphi(k)}$, so its distinct roots lie in $\mathbf{G}_m$. Proposition 8.1 maps them onto

$$\{(\lambda, C) \in X_4(\overline{\mathbf{Q}}) : \lambda \text{ is a primitive } k\text{-th root of unity}\}.$$

Indeed, surjectivity of the normalization supplies a preimage $T \in \overline{\mathbf{Q}}^{\times}$ of every point in this set, and then $\Phi_k(g(T)) = 0$; conversely every root of Γ_k supplies such a point. By (21), separability, and (22), both the source and target have $6\varphi(k)$ elements. The resulting surjection is therefore a bijection. It is Galois-equivariant because g and h are defined over $\mathbf{Q}$. $\square$

The absolute Galois group acts transitively on $Z(\Gamma_k)$ by Theorem 1.1. Lemma 8.2 and the separability of $\widetilde{\Delta}_{4k,4}$ therefore prove that $\widetilde{\Delta}_{4k,4}$ is irreducible for $k \geq 2$.

For $k = 1$, the table in [MV95, Section 3, Table 1] gives

$$\widetilde{\delta}_4(1, C) = (C^2 - 2C + 5)(C + 5)(C^3 + 9C^2 + 27C + 135).$$

Since $\widetilde{\delta}_1(x, C) = x^2 - 2x + C$ and $\widetilde{\delta}_2(x, C) = C + 4 - x$, the exact-period-4 factor is

$$\widetilde{\Delta}_{4,4}(C) = C^3 + 9C^2 + 27C + 135 = (C + 3)^3 + 108.$$

A rational root would be an integer and would make an integer cube equal to -108, which is impossible because $v_2(108) = 2$ is not divisible by 3. Thus the cubic is irreducible. The invertible linear change $C = 4c$ preserves irreducibility over $\mathbf{Q}$, completing the proof of Corollary 1.3.

Remark 8.3 (Scope of the rational parametrization)**.** *For the quadratic family, the normalizations of the multiplier curves X_m are rational exactly for $m \leq 4$. For $m = 1, 2$ this follows from the displayed elementary equations; period 3 is parametrized in [KMST25, Lemma 2.6], and period 4 is Proposition 8.1. For $m \geq 5$, Morton's genus formula and its quadratic-family corollary give genus at least two [Mor96, Theorem 13(b),(c) and the corollary following Theorem 14].[1] The local simplification used above also exploits fourth powers in residue characteristic two. Thus period 4 is the endpoint reached by the present implementation of the rational-parametrization method; this is not an obstruction to different methods at higher periods.*

[1]Morton's calculation is made under his hypothesis (H). Its clause asserting simple roots at multiplier 1 is justified there using the Douady–Hubbard description of hyperbolic components; see the discussion preceding hypothesis (H) in [Mor96].

9 Newton polygons and a denominator lemma

We first record precisely the local consequence of Newton polygons that will be used below. If L is a complete discretely valued field, its valuation is normalized by $v(L^\times) = \mathbf{Z}$, and the same letter v denotes a fixed extension to $\overline{L}$. We use the standard Newton polygon theorem over complete discretely valued fields [Ser79, Chapter II].

Lemma 9.1 (Newton-polygon denominator lemma)**.** *Let $P \in L[T]$ be monic with $P(0) \neq 0$. Suppose that the Newton polygon of P has an edge of horizontal length ℓ and slope*

$$-\frac{h}{d}, \qquad h \in \mathbf{Z}, \quad d \geq 1, \quad \gcd(h, d) = 1.$$

Then the slope factor belonging to that edge has degree ℓ, and every irreducible factor of the slope factor has degree divisible by d. In particular, if $\ell = d$, the slope factor is irreducible of degree d over L.

Proof. By the Newton polygon theorem over the complete, hence henselian, field L, the polynomial P has a factor $P_{h/d} \in L[T]$ whose roots, counted with multiplicity, are exactly the ℓ roots β of P for which

$$v(\beta) = \frac{h}{d}.$$

Thus $\deg P_{h/d} = \ell$.

Let $G \in L[T]$ be an irreducible factor of $P_{h/d}$ and let β be a root of G. Put $M = L(\beta)$ and let $e(M/L)$ be the ramification index. With the normalization $v(L^\times) = \mathbf{Z}$, the value group of M is

$$v(M^\times) = \frac{1}{e(M/L)}\mathbf{Z}.$$

Since $v(\beta) = h/d$ is in lowest terms, it follows that $d \mid e(M/L)$. The ramification index divides $[M : L] = \deg G$, so $d \mid \deg G$. If $\ell = d$, the degrees of the irreducible factors of $P_{h/d}$ are positive multiples of d whose sum is d; hence there is exactly one such factor. $\square$

10 A local quartic factor of the period-4 fiber

Let $\zeta \neq 1$ be a root of unity, put $K = \mathbf{Q}(\zeta)$, and write

$$\operatorname{ord}(\zeta) = 2^a m, \qquad m \text{ odd}, \qquad \zeta = \xi\eta,$$

where ξ has 2-power order and η has odd order. Recall that

$$F_\zeta(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (\zeta - 5)T^2 - 8T - 16.$$

Fix a prime $\mathfrak{p}$ of K above 2, write $K_\mathfrak{p}$ for the completion, normalize $v = v_\mathfrak{p}$ by $v(K_\mathfrak{p}^\times) = \mathbf{Z}$, and put $e = v(2)$.

When $m > 1$, choose the odd-order root of unity $\theta \in K$ used in Section 6, so that $\theta^4 = \eta$, and put $t_0 = \theta - 1$. The base irreducibility argument already proved that t_0 is a $\mathfrak{p}$-adic unit and that

$$\overline{F_\zeta(T)} = T^2(T - \overline{t_0})^4, \qquad v\big(F_\zeta(t_0)\big) = 1;$$

see (9) and Lemma 6.1.

Proposition 10.1 (Local quartic factor)**.** *For every root of unity $\zeta \neq 1$, the polynomial F_ζ has an irreducible factor of degree 4 over $K_\mathfrak{p}$.*

Proof. Suppose first that $m > 1$. Write

$$F_\zeta(t_0 + Y) = \sum_{j=0}^{6} b_j Y^j.$$

Lemma 6.1 gives $v(b_0) = 1$. Reducing the shifted polynomial and using (9), we obtain, in residue characteristic 2,

$$\begin{aligned}\overline{F_\zeta(t_0 + Y)} &= (Y + \overline{t_0})^2 Y^4 \\ &= Y^6 + \overline{t_0}^2 Y^4.\end{aligned}$$

Consequently

$$v(b_j) \geq 1 \quad (1 \leq j \leq 3), \qquad v(b_4) = 0, \qquad v(b_5) \geq 1, \qquad v(b_6) = 0.$$

Every point $(j, v(b_j))$ lies on or above the line $y = 1 - j/4$, and equality holds at $j = 0$ and $j = 4$. Hence the Newton polygon contains the edge

$$(0, 1) \longrightarrow (4, 0),$$

of length 4 and slope $-1/4$. Lemma 9.1 shows that the corresponding degree-four slope factor of $F_\zeta(t_0 + Y)$ is irreducible over $K_\mathfrak{p}$. Translating $Y = T - t_0$ preserves degrees and irreducibility, and therefore gives the required factor of F_ζ.

Now suppose that $m = 1$. Since $\zeta \neq 1$, its order is 2^a with $a \geq 1$. The recalled local cyclotomic facts give

$$e = v(2) = 2^{a-1}.$$

For $a = 1$ one has $\zeta = -1$ and $v(\zeta - 5) = v(-6) = 1$. For $a \geq 2$, the element $\zeta - 1$ is a uniformizer, and

$$\zeta - 5 = (\zeta - 1) - 4$$

is a sum of terms of respective valuations 1 and $2e > 1$; hence again $v(\zeta - 5) = 1$. It follows directly from the seven coefficients of F_ζ, listed from the constant coefficient to the leading coefficient, that their valuations are

$$(4e,\, 3e,\, 1,\, e,\, 2e,\, e,\, 0). \tag{23}$$

The lower convex hull of these seven points is

$$(0, 4e) \longrightarrow (2, 1) \longrightarrow (6, 0). \tag{24}$$

Indeed, the point $(1, 3e)$ lies strictly above the first segment because

$$3e > \frac{4e + 1}{2},$$

while the extension of that segment has negative ordinate at every integer abscissa $j \geq 3$, since at $j = 3$ its ordinate is $-2e + 3/2 < 0$. All the remaining coefficient valuations are nonnegative. On the other hand, the points with abscissas 0 and 1 lie strictly above the line through $(2, 1)$ and $(6, 0)$, whose ordinates there are $3/2$ and $5/4$, respectively, because $4e > 3/2$ and $3e > 5/4$. Finally, the points with abscissas $3, 4, 5$ lie strictly above the second segment, whose ordinates there are $3/4, 1/2, 1/4$, respectively. The two displayed slopes are

$$-\frac{4e-1}{2} \qquad \text{and} \qquad -\frac{1}{4},$$

and are strictly increasing. These observations also show directly that no omitted point lies below either segment. The second edge in (24) has length 4 and reduced slope $-1/4$. Lemma 9.1 therefore makes its slope factor an irreducible quartic over $K_{\mathfrak{p}}$. ☐

11 A sextic criterion for abelian disjointness

Let k be a number field and let

$$f(T) = \prod_{i=1}^{6}(T - r_i) \in k[T]$$

be an irreducible separable sextic. For each unordered partition $\{A, A^c\}$ of $\{1, \ldots, 6\}$ into two triples, put

$$s_A = \prod_{i \in A} r_i + \prod_{j \in A^c} r_j.$$

There are $\binom{6}{3}/2 = 10$ such partitions. Define the $3+3$ resolvent by

$$\mathcal{R}_f(S) = \prod_{\{A, A^c\}} (S - s_A). \tag{25}$$

The absolute Galois group of k permutes the ten unordered partitions; thus $\mathcal{R}_f(S) \in k[S]$.

Write k^{ab} for the maximal abelian extension of k in a fixed algebraic closure.

Theorem 11.1 (Abelian-disjointness criterion for sextics)**.** *Let α be a root of f and put $E = k(\alpha)$. Suppose that*

(i) *for some finite place v of k, the polynomial f has an irreducible quartic factor over k_v;*

(ii) *the resolvent $\mathcal{R}_f$ has no root in k.*

Then

$$E \cap k^{\mathrm{ab}} = k.$$

Consequently f remains irreducible over k^{ab}.

Proof. Set

$$D = E \cap k^{\mathrm{ab}}.$$

Because $D \subseteq E$, the extension D/k is finite. Moreover, k^{ab}/k is Galois with abelian Galois group. Every finite intermediate extension is therefore Galois and abelian over k; in particular, so is D/k. Since f is irreducible of degree six, $[E:k] = 6$, and the tower $k \subseteq D \subseteq E$ gives

$$[D:k] \mid 6.$$

It remains to exclude the three possibilities $[D:k] = 2, 3, 6$.

Suppose first that $[D:k] = 2$, and let τ be the nontrivial element of $\mathrm{Gal}(D/k)$. The minimal polynomial H of α over D is monic of degree $[E:D] = 3$ and divides f in $D[T]$. The polynomials H and $\tau(H)$ are distinct: otherwise H would have coefficients fixed by

τ, hence would lie in $k[T]$, contradicting the irreducibility of the degree-six polynomial f over k. Both are irreducible over D, so they are coprime. Their product is monic of degree six, divides f, and is fixed by τ; therefore

$$f = H\,\tau(H).$$

The roots of H form a triple $\{r_i : i \in A\}$ and those of $\tau(H)$ form the complementary triple. Thus

$$H(0) = -\prod_{i\in A} r_i, \qquad \tau(H)(0) = -\prod_{j\in A^c} r_j,$$

and consequently

$$-\operatorname{Tr}_{D/k}\big(H(0)\big) = \prod_{i\in A} r_i + \prod_{j\in A^c} r_j = s_A.$$

This is a root of $\mathcal{R}_f$ lying in k, contrary to hypothesis (ii). Hence $[D : k] \neq 2$.

Suppose next that $[D : k] = 3$. Let $q \in k_v[T]$ be an irreducible quartic factor of f, let β be a root of q, and choose a k-embedding

$$\sigma : E \hookrightarrow \overline{k_v} \qquad \text{with} \qquad \sigma(\alpha) = \beta.$$

Such an embedding exists because β is a root of the minimal polynomial f of α. Put $L = k_v(\beta)$, so $[L : k_v] = 4$. The compositum $k_v\sigma(D) \subset \overline{k_v}$ is canonically isomorphic to the completion D_w at the place w induced by σ, and it is contained in L because $D \subseteq E = k(\alpha)$.

Since D/k is Galois of prime degree three, the local degree $[D_w : k_v]$ is either 1 or 3. If it is 3, then the tower

$$k_v \subseteq k_v\sigma(D) \subseteq L$$

would force 3 to divide $[L : k_v] = 4$, which is impossible. If it is 1, then v splits completely in D and $\sigma(D) \subseteq k_v$. The minimal polynomial of α over D has degree $[E : D] = 2$; applying σ to its coefficients gives a quadratic in $k_v[T]$ having β as a root. This implies $[k_v(\beta) : k_v] \leq 2$, again contradicting $[L : k_v] = 4$. Thus $[D : k] \neq 3$.

Finally, suppose that $[D : k] = 6$. The inclusions $D \subseteq E$ and the equality of degrees give $D = E$. In particular, E/k is abelian Galois. With q, β, and σ as in the preceding paragraph, we have

$$k_v(\beta) = k_v\sigma(E),$$

because $E = k(\alpha)$. The field on the right is a completion of the Galois extension E/k, so its degree over k_v is the order of a decomposition group and therefore divides $[E : k] = 6$. Its degree is also 4, by the irreducibility of q. This contradiction excludes $[D : k] = 6$.

We have proved $D = k$. It remains to justify the assertion over the infinite extension k^{ab}. If f were reducible over k^{ab}, a nontrivial factorization would involve only finitely many coefficients. They would all lie in some finite intermediate extension

$$k \subseteq M \subseteq k^{\mathrm{ab}}.$$

The extension M/k is finite abelian Galois, and

$$E \cap M \subseteq E \cap k^{\mathrm{ab}} = k.$$

Thus $E \cap M = k$. Since M/k is Galois, the fields E and M are linearly disjoint over k. More explicitly, restriction identifies $\mathrm{Gal}(EM/E)$ with $\mathrm{Gal}(M/E \cap M) = \mathrm{Gal}(M/k)$; hence $[EM : E] = [M : k]$, and the tower formula gives

$$[M(\alpha) : M] = [E : k] = 6.$$

The minimal polynomial of α over M therefore still has degree six and equals f, contradicting the assumed factorization over M. Hence f is irreducible over k^{ab}. $\square$

12 The 3+3 resolvent

Put

$$P_X(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (X-5)T^2 - 8T - 16$$

so that $P_\zeta(T) = F_\zeta(T)$. Let $r_1, \dots, r_6$ be the roots of P_X in a splitting field over $\mathbf{Q}(X)$. If $A \subset \{1, \dots, 6\}$ has cardinality three, write $y_A = \prod_{i \in A} r_i$. For an unordered complementary pair $\{A, A^c\}$, put $s_A = y_A + y_{A^c}$. There are ten such pairs, and we define

$$\mathcal{R}_X(S) = \prod_{\{A, A^c\}} (S - s_A).$$

The coefficients are symmetric polynomials in the r_i with integer coefficients. The fundamental theorem on symmetric polynomials therefore gives $\mathcal{R}_X \in \mathbf{Z}[X, S]$.

Let C_X be the companion matrix of P_X, in a convention for which its characteristic polynomial is P_X, and put

$$Q_X(Y) = \det\Big(YI - \bigwedge^3 C_X\Big).$$

The eigenvalues of $\bigwedge^3 C_X$ are the twenty numbers y_A. Moreover,

$$y_A y_{A^c} = r_1 \cdots r_6 = -16, \qquad y_{A^c} = -\frac{16}{y_A}.$$

Consequently, for each complementary pair,

$$(Y - y_A)(Y - y_{A^c}) = Y^2 - s_A Y - 16,$$

and hence

$$Q_X(Y) = Y^{10} \mathcal{R}_X\left(Y - \frac{16}{Y}\right). \tag{26}$$

Exact expansion gives

$$\begin{aligned} \mathcal{R}_X(S) = S^{10} &+ 6S^9 + 4(X+35)S^8 + 2(X^2 - 6X + 389)S^7 \\ &+ (X^3 - 15X^2 + 667X + 5683)S^6 + 96(X^2 - 2X + 337)S^5 \\ &+ 64(X^3 - 14X^2 + 477X + 800)S^4 - 1536(X^2 - 21X - 368)S^3 \\ &+ 8192(X^2 + 35X - 48)S^2 - 65536(X - 16)(X + 4)S \\ &- 65536(X - 16)^2. \end{aligned} \tag{27}$$

Identity (27) is certified independently by the exact-arithmetic exterior-power computation described in Appendix A; no numerical specialization is used.

13 Absence of K-rational roots of the resolvent

Let $\zeta \neq 1$ be a root of unity, put $K = \mathbf{Q}(\zeta)$, and set $\mathcal{R}_\zeta = \mathcal{R}_X|_{X=\zeta}$. Fix a prime $\mathfrak{p} \mid 2$, normalize $v = v_\mathfrak{p}$ by $v(K_\mathfrak{p}^\times) = \mathbf{Z}$, and put $e = v(2)$. Write $\bar{x}$ for the reduction of an integral element x modulo $\mathfrak{p}$.

13.1 Orders with a nontrivial odd part

Write $\mathrm{ord}(\zeta) = 2^a m$ with $m > 1$ odd, and decompose $\zeta = \xi\eta$, where ξ has 2-power order and η has odd order. Since $\bar{\xi} = 1$ and reduction is injective on odd-order roots of unity, $\bar{\zeta} = \bar{\eta} \neq 1$.

Write $\mathcal{R}_\zeta(S) = \sum_{j=0}^{10} c_j S^j$. Reducing the unit part of each coefficient in (27) gives the following exact valuations:

$$\big(v(c_0), \dots, v(c_{10})\big) = e(16, 16, 13, 9, 6, 5, 0, 1, 2, 1, 0).$$

For completeness, the reductions that prove exactness are

$$\begin{aligned}\Big(c_j/2^{v(c_j)/e} \bmod \mathfrak{p}\Big)_{j=0}^{10} = \big(&\bar{\zeta}^2, \bar{\zeta}^2, \bar{\zeta}(\bar{\zeta}+1), \bar{\zeta}(\bar{\zeta}+1),\\ &\bar{\zeta}(\bar{\zeta}+1)^2, (\bar{\zeta}+1)^2, (\bar{\zeta}+1)^3, (\bar{\zeta}+1)^2,\\ &\bar{\zeta}+1, 1, 1\big).\end{aligned}$$

Odd rational-integer factors reduce to 1 and have therefore been suppressed; every displayed residue is nonzero.

The lower convex hull is

$$(0, 16e) \longrightarrow (6, 0) \longrightarrow (10, 0). \tag{28}$$

Indeed, after division by e, the differences between the valuations at abscissas $1, \dots, 5$ and the line from $(0, 16e)$ to $(6, 0)$ are

$$\frac{8}{3}, \quad \frac{7}{3}, \quad 1, \quad \frac{2}{3}, \quad \frac{7}{3},$$

all positive. From abscissa 6 onward the lower boundary is horizontal at height 0. The first six roots thus have valuation $8e/3$. The ramification index e is a power of two, so $8e/3 \notin \mathbf{Z}$; none of these six roots belongs to $K_\mathfrak{p}$.

It remains to exclude the four unit roots. Since $\gcd(4, m) = 1$, the map $x \mapsto x^4$ is an automorphism of the cyclic group $\langle\eta\rangle$. Let $\theta \in \langle\eta\rangle$ be the unique element satisfying $\theta^4 = \eta$, and put

$$s_0 = (\theta + 1)^3.$$

Since $\theta \neq 1$ and reduction is injective on odd-order roots of unity, $\bar{\theta} \neq 1$; hence s_0 is a $\mathfrak{p}$-adic unit. Only the coefficients of S^{10} and S^6 survive modulo $\mathfrak{p}$, and

$$\overline{c_6} = (\bar{\zeta} + 1)^3 = (\bar{\theta} + 1)^{12} = \bar{s}_0^4.$$

Thus

$$\overline{\mathcal{R}_\zeta(S)} = S^6 (S - \bar{s}_0)^4. \tag{29}$$

We shall use the following three coefficientwise congruences in $\mathbf{Z}[u]$:

$$\mathcal{R}_{u^4}\big((u+1)^3\big) \equiv 2u^3(u+1)^{27} \pmod 4, \tag{30}$$

$$\frac{\partial \mathcal{R}_X}{\partial X}\big((u+1)^3\big)\Big|_{X=u^4} \equiv (u+1)^{26} \pmod 2, \tag{31}$$

$$\mathcal{R}_{-u^4}\big((u+1)^3\big) \equiv 2u^3(u+1)^{26} \pmod 4. \tag{32}$$

They are verified by the exact-arithmetic certificate described in Appendix A, which checks every coefficient of the three differences.

We now prove

$$v\big(\mathcal{R}_\zeta(s_0)\big) = 1. \tag{33}$$

If $a = 0$, then $\zeta = \eta = \theta^4$ and $e = 1$; congruence (30) writes $\mathcal{R}_\zeta(s_0)$ as twice a unit plus an element of $4\mathcal{O}_{K_\mathfrak{p}}$, and proves (33). If $a = 1$, then $\zeta = -\eta = -\theta^4$ and $e = 1$, so (32) gives the same conclusion in the same way.

Suppose $a \geq 2$. Set $\delta = \eta(\xi - 1)$, so that $\zeta = \eta + \delta$ and $v(\delta) = 1$. Taylor expansion in the polynomial variable X gives

$$\mathcal{R}_{\eta+\delta}(s_0) = \mathcal{R}_\eta(s_0) + \delta\, \partial_X \mathcal{R}_X(s_0)|_{X=\eta} + \delta^2 A$$

for some $A \in \mathcal{O}_{K_\mathfrak{p}}$; its integrality follows because $\mathcal{R}_X(s_0)$ has coefficients in $\mathcal{O}_{K_\mathfrak{p}}$. Congruence (30) gives $v(\mathcal{R}_\eta(s_0)) = e \geq 2$, while (31) says that the displayed derivative is a unit. The linear term therefore has valuation one and every other term has valuation at least two, proving (33) without cancellation.

After the translation $S = s_0 + Y$, all coefficients remain integral, and equation (29) shows that the coefficients of Y, Y^2, Y^3 have positive valuation and that the coefficient of Y^4 is a unit. Together with (33), this puts the edge

$$(0, 1) \longrightarrow (4, 0)$$

in the Newton polygon. Its slope is $-1/4$ and its horizontal length is four. Lemma 9.1 therefore makes the corresponding degree-four factor irreducible. In particular, none of these four roots lies in $K_\mathfrak{p}$. Together with (28), this proves that $\mathcal{R}_\zeta$ has no root in K whenever $m > 1$.

13.2 Pure powers of two

First let $\zeta = -1$. Substitution in (27) gives $c_j = 2^{v(c_j)} u_j$, with the following odd unit parts u_j:

j	0	1	2	3	4	5	6	7	8	9	10
u_j	-289	51	-41	519	77	255	625	99	17	3	1,

and therefore

$$\big(v(c_0), \ldots, v(c_{10})\big) = (16, 16, 14, 10, 8, 7, 3, 3, 3, 1, 0).$$

The differences above the line from $(0, 16)$ to $(6, 3)$ at abscissas $1, \ldots, 5$ are

$$\frac{13}{6},\quad \frac{7}{3},\quad \frac{1}{2},\quad \frac{2}{3},\quad \frac{11}{6}.$$

The points at abscissas $7, 8, 9$ also lie strictly above the line from $(6, 3)$ to $(10, 0)$. Hence the lower convex hull is

$$(0, 16) \longrightarrow (6, 3) \longrightarrow (10, 0),$$

with slopes $-13/6$ and $-3/4$. The six roots belonging to the first edge have valuation $13/6$, and the four belonging to the second have valuation $3/4$. Neither valuation is integral. In particular, $\mathcal{R}_{-1}$ has no root in $\mathbf{Q}_2$.

Now suppose that ζ has order 2^a with $a \geq 2$. Put $\pi = \zeta - 1$ and $e = v(2) = 2^{a-1}$, so $v(\pi) = 1$. Substituting $X = 1 + \pi$ into the parenthesized coefficient polynomials in (27) gives

$$
\begin{aligned}
X^2 + 35X - 48 &= \pi^2 + 37\pi - 12, \\
X^2 - 21X - 368 &= \pi^2 - 19\pi - 388, \\
X^3 - 14X^2 + 477X + 800 &= \pi^3 - 11\pi^2 + 452\pi + 1264, \\
X^2 - 2X + 337 &= \pi^2 + 336, \\
X^3 - 15X^2 + 667X + 5683 &= \pi^3 - 12\pi^2 + 640\pi + 6336, \\
X^2 - 6X + 389 &= \pi^2 - 4\pi + 384, \\
X + 35 &= \pi + 36.
\end{aligned}
$$

In addition, $X - 16 = \pi - 15$ and $X + 4 = \pi + 5$ are units. Since $e \geq 2$, each displayed expression has a unique term of least valuation. It follows that

$$
\begin{aligned}
\big(v(c_0), \ldots, v(c_{10})\big) = (&16e, 16e, 13e + 1, 9e + 1, 6e + 2, 5e + 2, \\
&3, e + 2, 2e + 1, e, 0).
\end{aligned} \tag{34}
$$

For the first segment, the differences at abscissas $1, \ldots, 5$ are

$$
\frac{16e - 3}{6}, \quad \frac{7e}{3}, \quad \frac{2e - 1}{2}, \quad \frac{2e}{3}, \quad \frac{14e - 3}{6},
$$

all positive. At abscissas $7, 8, 9$, the valuations $e + 2, 2e + 1, e$ are strictly greater than the respective heights $9/4, 3/2, 3/4$ of the second segment. Thus the lower convex hull is

$$
(0, 16e) \longrightarrow (6, 3) \longrightarrow (10, 0),
$$

with slopes

$$
-\frac{16e - 3}{6}, \qquad -\frac{3}{4}.
$$

The integer $16e - 3$ is odd, and modulo three it is congruent to e. Since e is a power of two, $3 \nmid e$; hence

$$
\gcd(16e - 3, 6) = 1.
$$

The two slopes have reduced denominators six and four, respectively, and the corresponding horizontal lengths are six and four. Lemma 9.1 therefore shows that the two slope factors are irreducible of degrees six and four. In particular, $\mathcal{R}_\zeta$ has no root in $K_\mathfrak{p}$.

Combining the odd-part and pure-power cases proves that

$$
\mathcal{R}_\zeta(S) \text{ has no root in } K \qquad (\zeta \neq 1). \tag{35}
$$

14 Abelian stability

Proof of Theorem 1.2. Let α be a root of F_ζ and retain the notation $K = \mathbf{Q}(\zeta)$. Theorem 7.2 gives $[K(\alpha) : K] = 6$. Since K^{ab}/K is Galois, the standard degree formula gives

$$
[K^{\mathrm{ab}}(\alpha) : K^{\mathrm{ab}}] = [K(\alpha) : K(\alpha) \cap K^{\mathrm{ab}}].
$$

Thus the stated intersection equality is equivalent to irreducibility over K^{ab}. Proposition 10.1 supplies an irreducible quartic local factor of F_ζ. Equation (35) proves that the $3 + 3$ resolvent has no root in K. Theorem 11.1 now proves both assertions. □

15 Factorization over abelian base fields

For $k \geq 2$, the fixed-degree identity (3) gives

$$\Gamma_k(T) = \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} F_\zeta(T). \tag{36}$$

By Theorem 1.2, for every nontrivial root of unity ζ, the polynomial F_ζ is irreducible over $\mathbf{Q}(\zeta)^{\mathrm{ab}}$, and for every root α one has

$$\mathbf{Q}(\zeta, \alpha) \cap \mathbf{Q}(\zeta)^{\mathrm{ab}} = \mathbf{Q}(\zeta).$$

Fix a primitive k-th root ζ_k, put $K_k = \mathbf{Q}(\zeta_k)$, and let $B/\mathbf{Q}$ be a finite abelian extension. Set

$$D = B \cap K_k.$$

Because both $B/\mathbf{Q}$ and $K_k/\mathbf{Q}$ are Galois, restriction gives

$$\mathrm{im}\big(\mathrm{Gal}(\overline{\mathbf{Q}}/B) \longrightarrow \mathrm{Gal}(K_k/\mathbf{Q})\big) = \mathrm{Gal}(K_k/D).$$

The orbits on the primitive k-th roots therefore have cardinality $[K_k : D]$. Thus

$$\Phi_k(X) = \Psi_1(X) \cdots \Psi_r(X) \quad \text{in } B[X], \qquad r = [D : \mathbf{Q}], \quad \deg \Psi_i = [K_k : D].$$

Here the Ψ_i are the distinct monic irreducible factors over B. For each i, define

$$\Gamma_{\Psi_i}(T) = (-1)^{\deg \Psi_i} T^{2 \deg \Psi_i} \Psi_i(g(T)).$$

The fixed-degree identity $T^2(g(T) - \zeta) = -F_\zeta(T)$ gives

$$\Gamma_{\Psi_i}(T) = \prod_{\Psi_i(\zeta)=0} F_\zeta(T) \quad \text{and} \quad \Gamma_k = \prod_{i=1}^{r} \Gamma_{\Psi_i}.$$

Fix a root ζ of Ψ_i and a root α of F_ζ. Since $F_\zeta(0) = -16$, one has $\alpha \neq 0$, so the Laurent expression $g(\alpha)$ is defined. The extension BK_k/K_k is abelian, hence is contained in K_k^{ab}. Therefore F_ζ remains irreducible over BK_k, and

$$[BK_k(\alpha) : BK_k] = 6.$$

On the other hand, the relation $\zeta = g(\alpha)$ shows that $BK_k \subset B(\alpha)$, and in fact $B(\alpha) = BK_k(\alpha)$. Hence

$$[B(\alpha) : B] = 6[BK_k : B] = 6[K_k : D].$$

This is the degree of Γ_{Ψ_i}. Since $\Gamma_{\Psi_i} \in B[T]$ is monic and vanishes at α, it is the minimal polynomial of α over B and is irreducible. These irreducible polynomials are pairwise distinct: if an element α were a common root of Γ_{Ψ_i} and Γ_{Ψ_j}, then $g(\alpha)$ would be a common root of the coprime polynomials Ψ_i and Ψ_j. This proves the following theorem.

Theorem 15.1. *For $k \geq 2$ and every finite abelian extension $B/\mathbf{Q}$, the polynomial Γ_k has exactly*

$$[B \cap K_k : \mathbf{Q}]$$

irreducible factors over B, each of degree

$$6[K_k : B \cap K_k].$$

In particular,

$$\Gamma_k \text{ is irreducible over } B \quad \Longleftrightarrow \quad B \cap K_k = \mathbf{Q}.$$

All splitting after an abelian base change is therefore caused by the cyclotomic factor Φ_k; no sextic fiber splits further.

Corollary 15.2 (Factorization over the maximal abelian extension)**.** *For every $k \geq 2$, the complete irreducible factorization of Γ_k over $\mathbf{Q}^{\mathrm{ab}}$ is*

$$\Gamma_k(T) = \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} F_\zeta(T). \tag{37}$$

It has $\varphi(k)$ factors, all of degree six.

Proof. Since $K_k \subset \mathbf{Q}^{\mathrm{ab}}$ and $\mathbf{Q}^{\mathrm{ab}}/K_k$ is abelian, it is contained in K_k^{ab}. Each F_ζ is consequently irreducible over $\mathbf{Q}^{\mathrm{ab}}$. Distinct primitive roots give distinct polynomials, as their T^2 coefficients are $\zeta - 5$. Equation (36) now proves the assertion. □

16 Abelian factorization of the period-4 delta factors

Lemma 8.2 gives a $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$-equivariant bijection

$$\beta : Z(\Gamma_k) \longrightarrow Z(\widetilde{\Delta}_{4k,4}), \qquad \beta(\alpha) = h(\alpha).$$

Restricting the action to $\mathrm{Gal}(\overline{\mathbf{Q}}/B)$ preserves equivariance, so β identifies Galois orbits over B, including their cardinalities. Since the irreducible factors of a separable polynomial correspond exactly to these orbits, Theorem 15.1 transfers without requiring an explicit polynomial substitution.

Theorem 16.1. *For $k \geq 2$ and every finite abelian extension $B/\mathbf{Q}$, the period-4 delta factor $\widetilde{\Delta}_{4k,4}$ has exactly*

$$[B \cap K_k : \mathbf{Q}]$$

irreducible factors over B, each of degree

$$6[K_k : B \cap K_k].$$

Over $\mathbf{Q}^{\mathrm{ab}}$ its complete factorization consists of $\varphi(k)$ irreducible sextic factors. Indeed, restricting β to $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}^{\mathrm{ab}})$ still identifies the relevant orbits. More precisely, for every primitive k-th root ζ, the image under β of $Z(F_\zeta)$ is one orbit of cardinality six for $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}^{\mathrm{ab}})$. The corresponding irreducible sextic delta factor is

$$H_\zeta(C) = \prod_{\alpha \in Z(F_\zeta)} \bigl(C - \beta(\alpha)\bigr) \in \mathbf{Q}^{\mathrm{ab}}[C].$$

The same assertions hold for the unscaled factor $\Delta_{4k,4}(c)$, because $C = 4c$ is an invertible linear change over $\mathbf{Q}$.

Remark 16.2. *This is an orbit description; it does not identify the polynomial $H_\zeta(C)$ in the parameter variable with $F_\zeta(T)$.*

Corollary 16.3 (Coprime cyclotomic base change)**.** *Let $k \geq 2$ and $\ell \geq 1$ be integers, and let ζ_ℓ be a primitive ℓ-th root of unity. If $\gcd(k, \ell) = 1$, then both Γ_k and $\widetilde{\Delta}_{4k,4}$ remain irreducible over $\mathbf{Q}(\zeta_\ell)$.*

Proof. For coprime k and ℓ, one has $K_k \cap \mathbf{Q}(\zeta_\ell) = \mathbf{Q}$. Apply Theorems 15.1 and 16.1 with $B = \mathbf{Q}(\zeta_\ell)$. $\square$

Remark 16.4. *Corollary 16.3 is the period-4 analogue of the coprime cyclotomic-base result for period 3 in [KMST25, Theorem 1.2]. The full factorization theorems above go further in the period-4 setting by allowing every finite abelian base field and describing all resulting factors.*

17 The maximal abelian subfield of a parameter field

Corollary 17.1 (Maximal abelian subfield of a parameter field)**.** *Let $C = 4c$ be an exact-period-4 parabolic parameter in the scaled coordinate, with primitive k-th root multiplier ζ, where $k \geq 2$. Then*

$$\mathbf{Q}(C) \cap \mathbf{Q}^{\mathrm{ab}} = \mathbf{Q}(\zeta).$$

Thus $\mathbf{Q}(\zeta)$ is the largest subfield of $\mathbf{Q}(C)$ that is Galois abelian over $\mathbf{Q}$, and $C \notin \mathbf{Q}^{\mathrm{ab}}$.

Proof. Let α be the corresponding root of F_ζ, so $C = \beta(\alpha) = h(\alpha)$. The root correspondence is injective and $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$-equivariant. Consequently, for every $\sigma \in \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$,

$$\sigma(\alpha) = \alpha \quad \Longleftrightarrow \quad \sigma(C) = C.$$

The two elements have the same stabilizer, and therefore

$$\mathbf{Q}(C) = \mathbf{Q}(\alpha).$$

Since $\zeta = g(\alpha)$, the right-hand field already contains $K = \mathbf{Q}(\zeta)$ and equals $K(\alpha)$. The abelian-stability theorem gives

$$K(\alpha) \cap K^{\mathrm{ab}} = K.$$

Because $\mathbf{Q}^{\mathrm{ab}}/K$ is abelian, $\mathbf{Q}^{\mathrm{ab}} \subset K^{\mathrm{ab}}$. Moreover, $K \subset \mathbf{Q}(C)$ and $K \subset \mathbf{Q}^{\mathrm{ab}}$. The upper inclusion from the abelian-stability theorem and this lower inclusion give

$$\mathbf{Q}(C) \cap \mathbf{Q}^{\mathrm{ab}} = \mathbf{Q}(\zeta). \tag{38}$$

Thus the multiplier field is precisely the largest subfield $L \subset \mathbf{Q}(C)$ for which $L/\mathbf{Q}$ is Galois abelian. In particular, $C \notin \mathbf{Q}^{\mathrm{ab}}$: otherwise $\mathbf{Q}(C) \subset \mathbf{Q}^{\mathrm{ab}}$ would force $\mathbf{Q}(C) = K$, contradicting $[\mathbf{Q}(C) : K] = 6$. $\square$

When the multiplier is 1, the exact-period-4 parameters instead satisfy

$$p(C) = C^3 + 9C^2 + 27C + 135 = 0.$$

After $U = C + 3$, this becomes $U^3 + 108$. A rational root of this monic cubic would be an integer r satisfying $r^3 = -108$, which is impossible because -108 is not an integer cube. The cubic is therefore irreducible. Its discriminant is

$$-27 \cdot 108^2 = -2^4 3^9,$$

which is not a square. Its Galois closure consequently has group S_3, and the cubic field $\mathbf{Q}(C)$ is not Galois over $\mathbf{Q}$. Every finite subfield of $\mathbf{Q}^{\mathrm{ab}}$ is Galois over $\mathbf{Q}$, so these parameters also lie outside $\mathbf{Q}^{\mathrm{ab}}$. Since a cubic field has no nontrivial proper subfield, its largest subfield Galois abelian over $\mathbf{Q}$ is $\mathbf{Q}$. We conclude that no exact-period-4 parabolic parameter belongs to $\mathbf{Q}^{\mathrm{ab}}$.

18 Torsion Hilbert irreducibility context

For this interpretation one must use the finite cover of degree six

$$g : U \longrightarrow \mathbf{G}_m, \qquad U = \mathbf{G}_m \setminus g^{-1}(0).$$

Its degree is the degree of the rational function g, namely six. After writing the target coordinate as $X = g(T)$, the element T satisfies the monic equation $P_X(T) = 0$ with nonzero constant term; both T and T^{-1} are integral over $\overline{\mathbf{Q}}[X, X^{-1}]$, which also proves finiteness. The pullback condition means that the fiber product with $x \mapsto x^n$ is geometrically irreducible for every $n \geq 1$. Its function-field equation is

$$Y^n = g(T).$$

Over $\overline{\mathbf{Q}}(T)$, the function g is not a p-th power for any prime p. For $p \geq 3$, this follows from its pole of order two at $T = 0$. For $p = 2$, suppose $g = w^2$. Its only poles force

$$w = aT^2 + bT + c + dT^{-1}.$$

Comparison of the coefficients of T^4, T^3, T^{-1}, T^{-2} gives

$$a^2 = -1, \qquad b = a, \qquad d^2 = 16, \qquad cd = 4.$$

Thus $c^2 = 1$, whereas comparison at T^2 gives

$$-1 + 2ac = -4, \qquad ac = -\frac{3}{2},$$

so $(ac)^2 = 9/4$, contradicting $a^2c^2 = -1$. Hence g is not a square.

The field $\overline{\mathbf{Q}}(T)$ contains all roots of unity. The Kummer binomial criterion [Lan02, Chapter VI, §9, Theorem 9.1] in this setting states that $Y^n - q$ is irreducible if and only if q is not a p-th power for every prime $p \mid n$. In the general binomial criterion one must also exclude $q \in -4\overline{\mathbf{Q}}(T)^4$ when $4 \mid n$; here every element $-4b^4 = (2ib^2)^2$ is already a square, so the preceding square argument excludes this case as well. The preceding valuation and coefficient arguments therefore prove that $Y^n - g(T)$ is irreducible for every $n \geq 1$, which is precisely the pullback condition.

Zannier's cyclotomic Hilbert irreducibility theorem [Zan10, Theorem 2.1] says that a cover defined over the cyclotomic closure and satisfying this condition has irreducible fibers over that field outside a finite union of proper torsion cosets. For $\mathbf{G}_m$, such a union is a finite set of torsion points. Thus the theorem gives finitely many exceptional torsion fibers. By the Kronecker–Weber theorem [Was97], the cyclotomic closure of $\mathbf{Q}$ is $\mathbf{Q}^{\mathrm{ab}}$. The abelian-stability theorem determines them exactly: for $\zeta \neq 1$, the inclusion $\mathbf{Q}^{\mathrm{ab}} \subset \mathbf{Q}(\zeta)^{\mathrm{ab}}$ shows that F_ζ is irreducible over $\mathbf{Q}^{\mathrm{ab}}$, whereas

$$F_1(T) = (T + 2)(T^2 + 4)(T^3 - 2).$$

Thus $\zeta = 1$ is the unique exceptional torsion fiber for this cover.

A Exact symbolic certificates

The verification directory contains a dependency-free exact certificate using integer polynomial arithmetic. It constructs the companion matrix of P_X, constructs the matrix of its third exterior power from all 3×3 minors, and verifies every coefficient of the exterior-power resolvent identity and of the displayed resolvent formula. It also checks the three polynomial congruences and the parametrization identity (20) coefficient by coefficient. The program exits with a nonzero status on any failed equality or nonexact integer division; no floating-point computation or numerical specialization occurs.

From the directory containing this manuscript, run

```
python3 verification/verify_all.py
```

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